



Assessment of NO₂ concentrations over industrial state Jharkhand, at the time frame of pre, concurrent, and post-COVID-19 lockdown along with the meteorological behaviour: an overview from satellite and ground approaches

Shalini Priya¹ · Jawed Iqbal¹

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Abstract

Burning of fossil fuels in the form of coal or gasoline in thermal power plants, industries, and automobiles is a prime source of nitrogen dioxide (NO₂), a major air pollutant causing health problems. In this paper, spatio-temporal unevenness of NO₂ concentrations via both spaceborne Sentinel-5P and ground-based in situ data have been studied for the period of 2017–2021. Annual and seasonal distribution of TROPOMI-NO₂ depict consistency over the Jharkhand region, highlighting six hotspot regions. As compared to 2019, a notable dip of 11% in the spatial annual average TROPOMI-NO₂ was achieved in 2020, which were elevated again by 22% in 2021 as the lockdown gradually goes out of the picture. Among eight ground-monitoring stations, Tata and Golmuri stations always displayed a higher level of TROPOMI-NO₂ ranges up to 15.2×10^{15} molecules. cm⁻² and 16.9×10^{15} molecules. cm⁻² respectively, as being located in the highly industrialised district of Jamshedpur. A big percentage reduction of up to 30% in TROPOMI-NO₂ has been reported in Jharia and Bastacola stations in Dhanbad in the lockdown phase of 2020 compared to 2019. Good agreement between TROPOMI-NO₂ and surface-NO₂ has been achieved with $R = 0.8$ and $R = 0.71$ during winter and post-monsoon respectively. Among four meteorological parameters, TROPOMI-NO₂ was majorly found to be influenced by precipitation, having $R = 0.6$ – 0.8 for almost all stations. More advanced satellite algorithms and ground-based data may be used to estimate NO₂ in places where monitoring facilities are limited and thus can help in air pollution control policy.

Keywords Jharkhand · Surface-NO₂ · TROPOMI-NO₂ · Relative difference · GIS techniques · COVID-19-induced lockdown

Highlights

- The annual TROPOMI-NO₂ mapping for three consecutive years (2019–2021) was carried out for Jharkhand, India, an industrial and thermal power hub.
- TROPOMI NO₂ hotspots (high concentration) appeared over Dhanbad, Jamshedpur, Koderma, and the periphery of Godda, Sahibganj, and Pakur.
- Compared to the pre-COVID period (2019), the COVID period (2020) shows a reduction of 11% in the spatial annual average TROPOMI-NO₂ level with a contrast of 22% increment in the post-covid period (2021).
- Precipitation is the most significant meteorological parameter in deciding the TROPOMI-NO₂ level at eight ground-monitoring stations.
- Good seasonal consistency between TROPOMI-NO₂ and ground-NO₂ has been achieved with $R = 0.8$ and $R = 0.71$ ($p < 0.01$) during winter and post-monsoon respectively.

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Extended author information available on the last page of the article

Introduction

Pollution is a big threat to human existence, air pollution in particular claims around seven million human lives and poses a potential threat in long-term irreversible health risks (WHO, 2020). The South Asian countries such as China and India account for 36.5% of the world's total population, which is at risk (UN, 2015). Ever increasing population coupled with the fast-growing economy, rapid and unplanned steps towards urbanisation, socioeconomic development, and transportation has resulted in a remarkable rise in air pollution, especially in Northern India that are highly industrialised and populated area.

Industrialisation, vehicular movement, and thermal power plants are identified as the significant anthropogenic contributor to NO_x emissions by numerous studies such as V. Singh et al. (2020) and Hilboll et al. (2017). Other sources of

NO_x emission studied by various authors are forest fires (Hilboll et al., 2017), lightning (IPCC, 2007), soil (Ghude et al., 2010), solid biomass burning for heating and cooking, and stubble burning by farmers, to name a few. Nowadays, the study of oxides of nitrogen NO_x ($=\text{NO}+\text{NO}_2$) has fascinated the researchers owing to its role as a precursor in the genesis of tropospheric ozone (Monks et al., 2015). The presence of NO_x in the troposphere has resulted in adverse health effects (World Health Organization, 2003; Schraufnagel et al., 2019) due to climate change (Lin et al., 2015), acid rain formation, and secondary aerosol formation as well. The variation in tropospheric NO_x concentration is dependent upon the variation in seasons, climatic conditions, social status, and energy consumption (Zhou et al., 2012). In the Indian context, transportation is the major source of NO_x emission followed by thermal power plants, industries, and biomass burning (Garg et al., 2001).

In December 2019, COVID-19 was declared as a highly contagious disease (Chen et al., 2020) and later declared as a pandemic in year 2020 by WHO (Zhu et al., 2020; WHO 2020). The Government of India announced the COVID-19-induced lockdown at the midnight on 24th March for 21 days (25th March–14th April 2020) as phase 1, but as the situation was still not under control, extending it to May 31, 2020, continued in four different phases. Owing to the COVID-19 lockdown, various restricted anthropogenic activities such as vehicular, industrial, and other commercial energy-consuming activities (Jain & Sharma, 2020) negatively affected the country's social and economic growth, on the contrary, on pollution front, it positively affected the country, by lifting-up the air and water quality, reducing noise level and land surface temperature too (Mandal & Pal, 2020). Some recent studies adopting satellite and in situ-based observations have disclosed that a significant drop in NO_2 concentrations happened during the lockdown phase across the globe.

Ground observation data in Bangalore showed a 87% drop in the NO_2 level (Navinya et al., 2020) and 52.68% in New Delhi (Mahato et al., 2020). An identical finding was observed by Xu et al. (2020) for three major cities of China for NO_2 concentrations with a decrease of 35.6% in February and March 2020 analysed against the same period during 2017–2019, whereas an 18% decline in NO_2 concentration was all over India (S. Sharma et al., 2020b). The decrease in NO_2 concentration during this time was also observed in the USA (Berman & Ebisu, 2020) and two major cities of Spain (Madrid and Barcelona) (Baldasano, 2020). According to a study by Nguyen et al. (2022), a reduction of 75.8, 55.9, 21.4, and 60.7% for NO_2 , $\text{PM}_{2.5}$, O_3 , and SO_2 concentrations respectively was found in Hanoi, Vietnam. Filonchik and Peterson (2020) also identified a drop of 9%, 77%, 31.3%, 60.4%, and 3% for $\text{PM}_{2.5}$, PM_{10} , SO_2 , NO_2 , and CO respectively through the lockdown period analysed against the same period in 2019.

Satellite observation has also played a crucial role in assessing tropospheric NO_2 concentration as tropospheric vertical column density (TVCD). Cao and Han (2022), using OMI- NO_2 data, observed that owing to the COVID-19 epidemic, a momentary disruption in socio-economic affairs is observed in China. While comparing the findings for the whole of India (S. Sharma et al., 2020b) deploying OMI tropospheric- NO_2 (Biswal et al., 2020), the results are comparative. The decreasing trend in NO_2 concentration was observed along with SO_2 , $\text{PM}_{2.5}$, and PM_{10} in seven cities of Rajasthan because of the lockdown (Sharma et al., 2020a, b). Biswal et al. (2021) observed a NO_2 reduction of approx. 20–40% in urban and approx. 15–25% in rural regions of India using satellite data for the period 25 March–3 May 2020.

Filonchik et al. (2021) also identified a stark change in Poland's air quality during lockdown as analysed against the same period in 2018 and 2019. Whereas, Hoang et al. (2021) came up with a detailed analysis of NO_2 concentrations covering most part of the world and the impact of behavioural change among people owing through the lockdown. On the other hand, Kumar et al. (2022) applied interpolation techniques of ArcGIS for air quality mapping in Mumbai, India. In contrast to the decreasing trend in NO_2 concentration worldwide, the continued trend in NO_2 level in the plain of China was reported by Wang et al. (2020) and three cities (Johannesburg, Durban, and Cape Town) of South Africa (Matandirotya & Burger, 2022) which deny a linear correlation between emission and pollution level.

Most of the recent studies have either been investigated for a few weeks or considered the few months of lockdown, pre-lockdown phases of the years 2019 and 2020. Authors have not come across any literature on understanding the NO_2 variability and its behaviour in post COVID-19 lockdown year (2021) and their comparison with the COVID-19-induced lockdown year (2020) and pre-COVID-19 lockdown year (2019).

Considering its sources and deadly consequences on health and the environment as mentioned above, an attempt has been made to carry out exploratory, descriptive, and explanatory analysis towards our interested domain in a better way which is yet less explored, despite it being the 3rd largest industrialised state of India. In this research, Jharkhand is explored in detail and is reported as it is and tried to explain further why things happened or why the situation exists with the support of studies that have been done.

Staying focussed on our target, research has been designed to analyse and interpret to present a clear picture of the spatial distribution of NO_2 and to identify exact information for its temporal variability in concentration, in the light of before pandemic-induced lockdown, during the lockdown, and after lockdown scenarios based on satellite observations and ground-based measurements as well. A key

asset of this discussion would be to determine the role of meteorological behaviour on the TROPOMI-NO₂ level at eight JSPCB monitoring stations and the analyses of percentage change in TROPOMI-NO₂ in all the 24 different districts and the above-mentioned stations too of Mega-mining state, Jharkhand falling under the same time zone. Present research thus aimed to provide a complete understanding in identifying the pollution level existing in Jharkhand, which will facilitate future studies to acknowledge the right places for improvements and to assess potential health hazard.

Materials and methods

Study area

Jharkhand state (Fig. 1) was notified 28th state of India on 15 November 2000, after being split from Bihar. It is the hub of industrial and mining activities, stretching vast from 21°59'N to 25°18'N latitude and 82°52'E to 87°54'E longitude in the Northern part of India. The state is spread in a total area of 79,714 km². Being the resort of many minerals and the powerhouse for mining activities, Jharkhand is positioned as the 3rd largest industrialised state of India. It is landlocked by the other Indian states of West Bengal (east), Uttar Pradesh (north-west), Chhattisgarh (west), Bihar (north), and Odisha (south) (source: <https://www.jharkhand.gov.in/>).

Data used

Three types of data from different agencies have been taken into account for this research to be carried out viz., 3 years (2019–2021) Sentinel-5p TROPOMI-NO₂ data, surface-NO₂ data (2017–2021) and meteorological data (2019–2021). Explanations supporting the selection of the mentioned variables for our present discussion are the following. NO₂ is defined as a criteria pollutant and Jharkhand is loaded with all the active major sources that emit NO₂, so, being an aware and responsible citizen, it is required to track its presence regularly for the betterment of our environment. Still, Jharkhand is lacking an adequate number of monitoring stations, so to compensate for that, the application of satellite observation as TROPOMI-NO₂ is carried out to monitor its concentration.

As the prevailing meteorological environment put its remarkable influence on the excellence of ambient air, which decides its lifetime in the atmosphere, it is regarded as one of the supreme drivers for deciding the goodness of air quality. As such, high winds dilute the pollutant towards the downwind direction whereas calm winds can stagnate pollutants in a smaller area. Temperature and relative humidity are favourable meteorological parameters in the formation

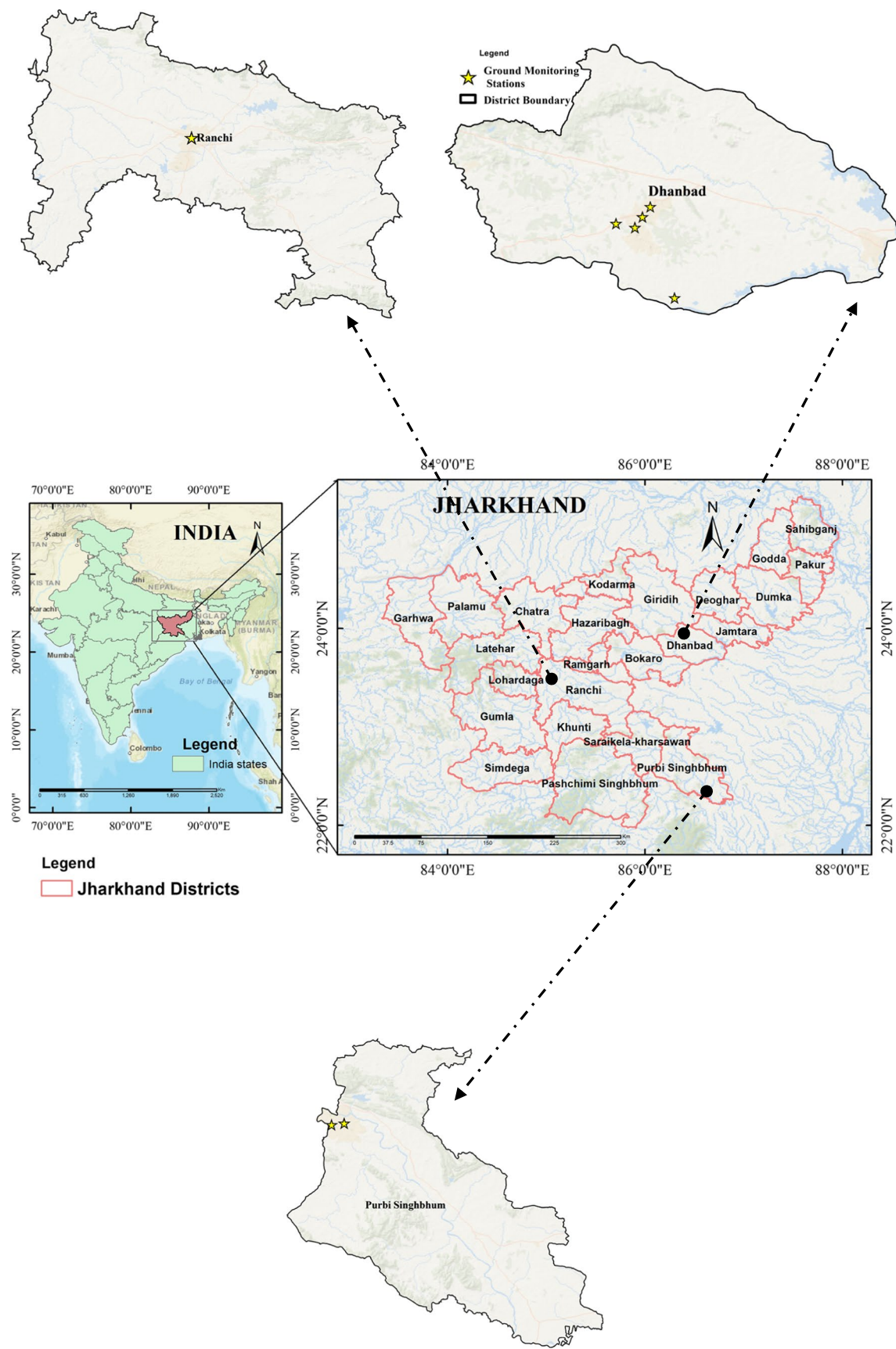
of secondary pollutants such as ozone and nitrate aerosols along with solar radiation and other chemicals present in the atmosphere. The longer lifetime of NO₂ in the atmosphere is determined by lower temperatures which restrict the production of OH and RO radicals whereas precipitation helps in lowering the pollution level through scavenging. Much more details of the data and its sources have been discussed below.

Satellite data

As of having advanced algorithms and improved sensitivity of Sentinel-5p regarding NO₂ detection at the lower atmosphere, its NO₂-tropospheric vertical column density data (TVCD) have been utilised for analysing and monitoring. On October 13, 2017, European Space Agency (ESA) launched Sentinel-5 Precursor, a satellite to track worldwide air pollution using its onboard sensor named TROPOMI (tropospheric monitoring instrument) (Potts et al., 2021). TROPOMI, a hyperspectral spectrometer, has the capabilities to sense ranging from ultraviolet to shortwave infrared portions of electromagnetic spectrum (Veefkind et al., 2012). It measures the column concentrations of various atmospheric gaseous and particulate pollutants, e.g. CO, and NO₂ (Zheng et al., 2019; Omrani et al., 2020) that encompass the full depth of the atmosphere.

The total columnar NO₂ is retrieved using the TROPOMI's spectrometer backscattered solar radiation which is the ratio of slant column density of NO₂ and total air mass. The retrieval of the total NO₂ column density was done using Level-1b radiance and irradiance spectra, measured by TROPOMI using the differential optical absorption spectroscopy (DOAS) approach (Platt & Stutz, 2008). Stratospheric and tropospheric NO₂ column density is separated using data assimilation system information based on the TM5-MP chemical transport model.

It is followed by the conversion of the tropospheric and stratospheric slant column densities into vertical column densities. The conversion is carried out by multiplying it by an appropriate air mass factor which depends on altitude and data on the vertical distribution of NO₂ from the TM5-MP chemical transport model. Owing to global coverage with temporal resolution for 1 day, and spatial resolution of 0.01° (Cooper et al., 2022), Prakash et al., 2021, this data helped in interpreting major pollutants over large Indian cities through the COVID-19 lockdown and found a decrease in concentrations specifically for NO₂ and SO₂. For a better performance and bigger span of study than the near real-time (NRTI) assets, offline (OFFL) Sentinel-5P TROPOMI-NO₂ products have been retrieved for the present analysis (Soleimany et al., 2021). Subsequently, to identify its variation on three different time scales, annual, seasonal and monthly TROPOMI-NO₂ products were extracted and downloaded using an open-source cloud-based platform Google Earth Engine



◀**Fig. 1** Study area map of Jharkhand State along with 24 administrative districts and geographic locations of eight JSPCB NO₂-ground-monitoring stations. These eight NO₂-ground-monitoring stations are picturized separately in their respective district boundary for their clear visibility. This map was generated using the Arc-map tool (ver. 10.4)

(GEE) (Source: <https://code.earthengine.google.com/>), by computing mean values of each day for the three consecutive years 2019, 2020, and 2021.

In situ NO₂ data

Assessment and analysis of NO₂ variation were carried out with the aid of Surface-NO₂ data for 8 ground-monitoring stations (Table S1) has been accessed from the Jharkhand State Pollution Control Board (JSPCB) office for the period of 2017 to 2021. Of these eight selected ground-monitoring stations, 5 are in Dhanbad, 2 in Jamshedpur, and 1 in Ranchi, which are highly urbanised and industrialised districts and are represented separately in their respective shapefiles in (Fig. 1) for a visual depiction of this distribution. Ground-based data was collected from Jharkhand State Pollution Control Board (JSPCB); a total of 104 samples were collected in a year by the agency, on a 24-h basis, twice a week. Among eight monitoring stations (of JSPCB), Golmuri [J], Kusunda [D], Bastacola [D], Jharia [D], and Sindri [D] belong to the industrial area whereas Albert Ekka [R] represents a blend of commercial and residential locales, Tata [J] again is a combination of commercial and residential areas whereas City Plaza [D] is a residential area only.

Meteorological data

Temperature(°C), wind speed (m/s), relative humidity (%), and precipitation (mm/day) were considered meteorological parameters for the period of 2019–2021 on a mean monthly basis from the open-source NASA website (source: <https://power.larc.nasa.gov/data-access-viewer/>), to perform an interpretation regarding its effect on TROPOMI-NO₂ behaviour for eight NO₂ ground-monitoring stations during the study period.

Data-pre-processing method

The steps involved in data acquiring and processing are summarised in a simple flow diagram (Fig. 2). Starting with annual, seasonal, and monthly mean TROPOMI-NO₂ data, they were downloaded and processed for the period of 2019 to 2021 using GEE (Gorelick et al., 2017). The researchers find the mean value of the desired parameter for the required time zone and region of study, process data using the code editor tab on the GEE website, and finally export and display

them in that order. For this purpose, we used the GEE website, where code has been written in the code editor tab by the researchers for data processing, which sets out finding the mean value of the desired parameter for the required time zone and region studied, and finally for displaying and exporting them accordingly.

For the post GEE processing of downloaded images such as reprojection into WGS 1984, unit conversion and clipping, ArcGIS software (ver. 10.4) was used and finally mapped into the presentable mode for further analysis. Thereafter, the collected mean surface-NO₂ and mean meteorological data for their respective stations were processed and arranged accordingly to get it into the desired form for spatial and temporal collocation with mean TROPOMI-NO₂ data. At the end, GPS locations of all 8 monitoring stations were used to extract the TROPOMI-NO₂ value for those stations out of processed and finalised maps for carrying out further analysis and interpretation using a statistical tool for a graphical representation.

Analysis methodology

By using the following equation (Biswal et al., 2021), the relative difference (%) in TROPOMI-NO₂ levels were calculated and analysed for each of the desired domain and time period:

$$\text{Relative Difference (\%)} = \frac{(\text{Present NO}_2 \text{ value} - \text{Previous NO}_2 \text{ value})}{\text{Previous NO}_2 \text{ value}} \times 100$$

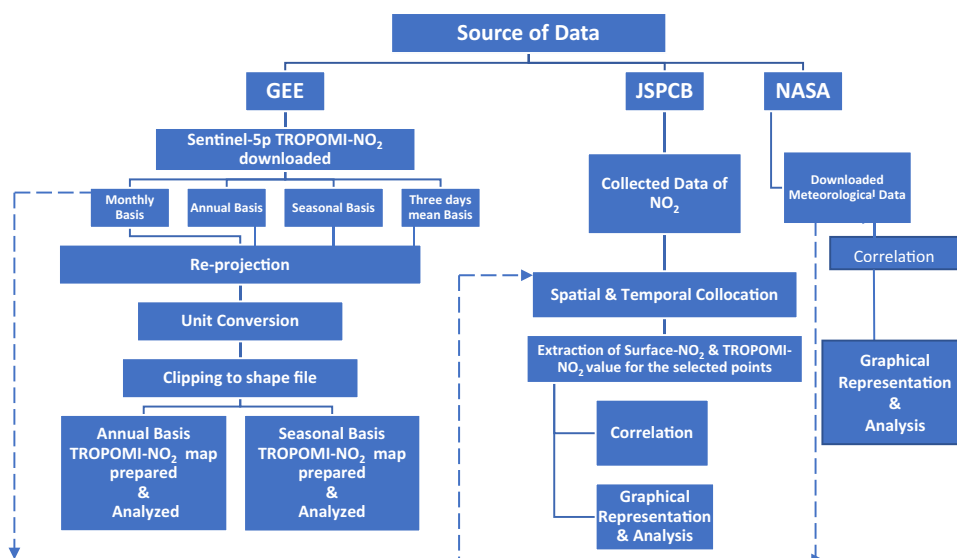
Results and discussion

The assessment of spatiotemporal analysis of NO₂ concentration will help identify the sources of NO₂ emissions and in constructing a strong strategy to minimise its concentration in ambient air. Hence, in the upcoming sections, a detailed analysis of NO₂ variability has been done for different time zones.

Year-wise spatial distribution of TROPOMI-NO₂ over Jharkhand

This section has employed annual mean satellite observations, to present and discuss the spatial and temporal view of TROPOMI-NO₂ for the three consecutive years 2019, 2020, and 2021 as pre-COVID, COVID, and post-COVID-led lockdown periods respectively. The annual TROPOMI-NO₂ distribution over the three consecutive years (2019–2021) was found to be consistent over the Jharkhand state, i.e. elevated TROPOMI-NO₂ value has been observed consistently for Dhanbad, Jamshedpur, Kodarma, and the periphery of Godda, Sahibganj, and Pakur. The aforementioned regions

Fig. 2 Flowchart of step-by-step representation of adopted working methodology



exhibiting a high TROPOMI-NO₂ value are identified and termed as the TROPOMI-NO₂ hotspot (high concentration) regions, dominated by many large-scale industries, namely petroleum refineries and thermal power stations.

Also, Jharkhand is surrounded by other industries and thermal- power plants(TPPs) of their nearby states Bihar, Uttar Pradesh, and West Bengal (Bandyopadhyay et al., 2021), which are ultimately subsidising hotspots for boosting TROPOMI-NO₂ levels. Ghude et al. (2008) found TROPOMI-NO₂ hotspots majorly to be allocated over industrial zones and thermal power plant centres. In the central Indian state of Jharkhand, the dominant sources of tropospheric NO₂ are thermal power plant emissions and the captive power plants of industries (Prasad et al., 2006). However, most of the other parts of Jharkhand (including western and

the southwestern regions of Jharkhand) consistently showed lower TROPOMI-NO₂ during 2019–2021.

The sequence of the spatial distribution of annual mean columnar TROPOMI-NO₂ for the whole of Jharkhand considering before (2019), during (2020), and after (2021) COVID-19 pandemic–led lockdown year is represented in Fig. 3. Spatial annual average TROPOMI-NO₂ over Jharkhand was found to be $2.34 \pm 0.82 \times 10^{15}$ molecules.cm⁻², $2.08 \pm 0.73 \times 10^{15}$ molecules.cm⁻², and $2.53 \pm 0.95 \times 10^{15}$ molecules.cm⁻² for the years 2019, 2020, and 2021 respectively. The difference in spatial annual average TROPOMI-NO₂ between 2019 and 2020 is 0.26×10^{15} molecules.cm⁻², whereas between 2020 and 2021, it is 0.45×10^{15} molecules.cm⁻². Accordingly, the standard deviation between 2019 and 2020 was found to be

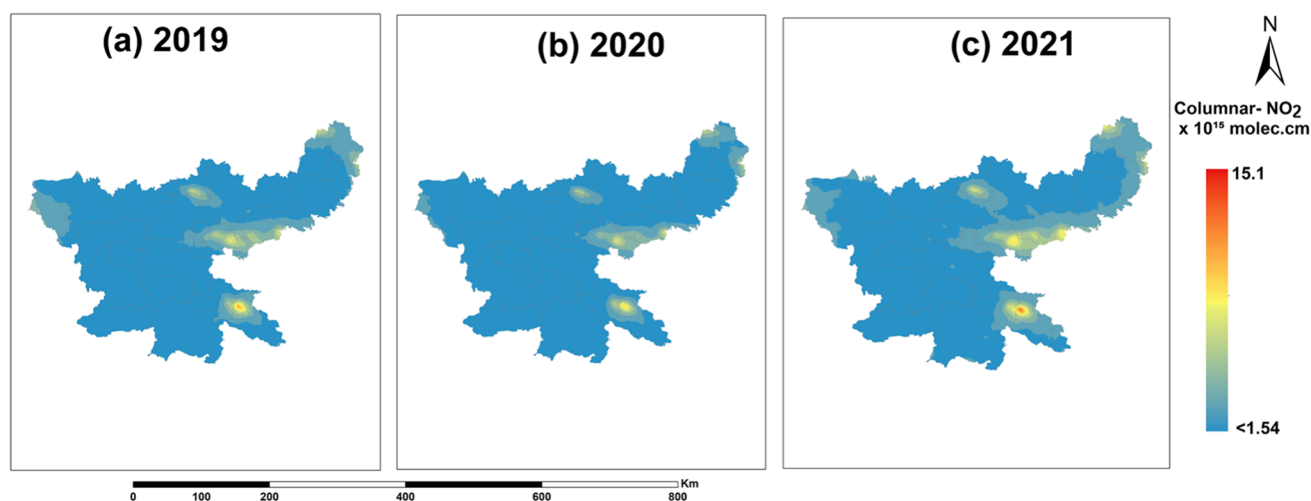


Fig. 3 Spatio-temporal distribution of annual mean TROPOMI-NO₂ for Jharkhand state during study period: **a** 2019 (pre-COVID-induced lockdown), **b** 2020 (COVID-induced lockdown), and **c** 2021 (post-COVID-induced lockdown)

0.09×10^{15} molecules. cm^{-2} , and between 2020 and 2021, it is 0.22×10^{15} molecules. cm^{-2} . The maximum and minimum measurements of average annual TROPOMI- NO_2 for each of the three considered years were also reported in Table S2.

Among the three scenarios, the COVID-induced lockdown period (2020) exhibited the minimum concentration of spatial annual average TROPOMI- NO_2 level, whereas the pre-COVID-induced lockdown period (2019) achieved moderate and post-COVID-induced lockdown period (2021) got the maximum spatial annual average TROPOMI- NO_2 level. A reduction of 11% in the spatial annual average TROPOMI- NO_2 level has been observed during the COVID period while comparing it with the pre-COVID period (2019) with a contrast of 22% increase in spatial annual average TROPOMI- NO_2 level in the year 2021 against the year 2020 level (Table S2). Soni et al. (2021) research is in concurrence with the findings of this research where the mean tropospheric NO_2 column is noticed to have $2.32 \pm 0.15 \times 10^{15}$ molecules. cm^{-2} during 2009–2019 and $1.95 \pm 0.30 \times 10^{15}$ molecules. cm^{-2} in the year 2020. A 16% decline in tropospheric columnar load of NO_2 in 2020 was observed as compared to 2009–2019 over India.

For better visualisation and understanding, the observed difference and relative difference obtained with respect to their previous years are shown in Fig. S1 a and b, which highlights a clear change in TROPOMI- NO_2 level for the region studied. Yellow, in the colour palette which was utilised in Fig. S1 a and b, indicates no changes occurred. The reduction in atmospheric NO_2 level due to the lockdown was attributed to lowered momentum in major anthropogenic functions, namely transportation and function of industrial and commercial enterprises. It is reflected that overall reductions are not so stunning, as they are spatial averaged for the whole of Jharkhand and might be the aggregate effect of power generation, biomass burning, and fossil fuels burning, and the activities vital to people's basic needs remained uninterrupted. Also, the major operations of the city that were still functioning continued to add to the atmospheric NO_2 emissions.

Hence, in the later part of the following section, a discussion on district-specific and monitoring station's surrounding environment-specific reductions could find more significance. The increase in the NO_2 level in the post-COVID regime reflects the normalcy attended by the region which restored the NO_2 concentration to pre-COVID time due to extensive emissions as a result of the restoration of all activities to normal levels.

Month-wise spatial distribution of TROPOMI- NO_2 over Jharkhand

To assess the monthly pattern in TROPOMI- NO_2 level, monthly mean satellite data were taken into consideration for the year 2019 to 2021. The COVID-induced lockdown period (2020) exhibited lower spatial monthly mean TROPOMI- NO_2 throughout 12 months as compared with the same period for the other two preceding and successive years (2019 and 2021). As discussed earlier, this decline in TROPOMI- NO_2 during the COVID-induced lockdown period (2020) is mainly contributed by social and travel constraints which catalysed the closing down of all the businesses that take in industries, mining, restaurants, transport, markets, tourism, construction, etc., which effectively contributed to reduced NO_2 emission, predefined essential services being exempted.

The monthly variation of spatial mean TROPOMI- NO_2 for the whole Jharkhand is represented in Fig. S2, which pronounced the presence of two peaks during the study period: one in June (2019) and the other in March (2021) having 2.88×10^{15} molecules. cm^{-2} and 3.56×10^{15} molecules. cm^{-2} TROPOMI- NO_2 levels respectively. Whereas no such peak has been captured during the COVID-19 year. These two peaks contribute 11.09% and 12.60% of the total in the corresponding years 2019 and 2020 as represented in Fig. S3. It could be assigned to the cumulative effect of the meteorological condition and an increased traffic movement post summer vacation. Variation in columnar NO_2 level due to meteorology is reported by Goldberg et al. (2020) which has been studied in the spring of 2019 and 2020 for the USA. The reasons for the existence of the peak in March 2020 was the holiday for the festival Holi and after the gradual end of a long period of COVID-19-induced lockdown, the over-excited public started their movement to their workplace and other locations.

Monthly NO_2 changes (Fig. S2) highlight that spatial average TROPOMI- NO_2 are always found to be high all through the months of the post-COVID-induced lockdown year (2021) except June as compared to the same months of the other two considered years 2019 and 2020. It could be one of the reasons that the public was busy coping with COVID-19, as the situation was alarming, and hence, they got off the track to reduce air pollution. The heat map of monthly averaged spatial mean TROPOMI- NO_2 for the whole of Jharkhand during 2019–2021 is represented in Fig. S4, which is compared and examined using three colour scales. The red-coloured cells show high TROPOMI- NO_2 , which are most of the months of winter, pre-monsoon, and post-monsoon, white-coloured cells represent moderate value whereas green-coloured cells depict low TROPOMI- NO_2 which mainly captures the monsoon months of all the considered years.

Owing to the positive impact of the COVID-19 lockdown, the spatial monthly average TROPOMI-NO₂ of Jharkhand state witnessed a drop of up to 1.90×10^{15} molecules.cm⁻² during March (2020) and resulting in a shift to the green-coloured cell. Overall, the highest spatial monthly average TROPOMI-NO₂ is found in March (2021) having 3.56×10^{15} molecules.cm⁻², and the minimum is observed in August (2020) measuring 1.36×10^{15} molecules.cm⁻². In a quick glance, this technique will help us swiftly interpret the available data is all about.

Season-wise spatial distribution of TROPOMI-NO₂ over Jharkhand

Winter (Jan–Feb), pre-monsoon (Mar–May), monsoon (Jun–Sep), and post-monsoon (Oct–Dec) are the four seasons classified by the Indian meteorological department (IMD) (source: <https://mausam.imd.gov.in/>). TROPOMI-NO₂ maps averaged over the period 2019–2021 figure out the considerable seasonal characteristics of its spatio-temporal spread, which are presented in Fig. S5a–d. Almost similar spatial patterns were observed in all the three seasons whereas a little lower TROPOMI-NO₂ was observed during monsoon. Nowadays, air pollution is not only a winter event in the country. The monitored seasonality of TROPOMI-NO₂ is found to be affiliated with the meteorological changes and seasonal cycles of anthropogenic emissions (Ghude et al., 2008). Among the four seasons, winter experiences the highest spatial mean TROPOMI-NO₂ $2.57 \pm 1.14 \times 10^{15}$ molecules.cm⁻², followed by the pre-monsoon $2.48 \pm 0.81 \times 10^{15}$ molecules.cm⁻², post-monsoon $2.21 \pm 0.99 \times 10^{15}$ molecules.cm⁻², and monsoon $1.83 \pm 0.47 \times 10^{15}$ molecules.cm⁻² shown in (Fig. S6).

Meteorological states in winter are additionally unfavourable for pollutant dispersion and dilution as compared to the other three seasons (Yang et al., 2019). An increase in adverse meteorology such as having a shallower planetary boundary layer (PBL), less UV radiations, and stable wind contribute towards a surge in the pollution level during winter. Besides that, during winter, NO₂ enhancement is also enriched by high biomass fuel consumption for heating homes (Ul-Haq et al., 2015), where OMI-tropospheric NO₂ over the South Asia regions were analysed during the past decade 2004–2015.

As per the Census of India 2011, 33 million is the population of Jharkhand, out of which 76% of the people are living in rural areas and the remaining 24% in urban areas, whereas the literacy rate in Jharkhand state is stated around 56–67.63% (Census, 2011). Hence, it helps to clarify that not many more people were aware of the harmful effects of biomass burning for domestic heating. Relatively high values in TROPOMI-NO₂ during pre-monsoon and post-monsoon are mainly accompanied by the release of NO₂ from significant

biomass and large-scale open-field burning of crop residue through the wheat-rice rotation period (Ul-Haq et al., 2015).

Agriculture is the foremost economic activity in Jharkhand by which about 63% of the rural population of the state is employed (International Institute For Population Sciences, 2016). During pre-monsoon, elevated TROPOMI-NO₂ are also partially added by intensified soil emissions owing to fertilisers being used in rice fields (Ul-Haq et al., 2015) (Ghude et al., 2008). Low TROPOMI-NO₂ during monsoon season is predominantly associated with excessive rainfall, strong winds, and elevated humidity. Prior studies have examined and found a remarkable negative correlation between NO₂ and rainfall (Martin, 1984). A 3% decrease is noted during pre-monsoon in TROPOMI-NO₂ as compared to their last winter season (Table S3). Similarly, a reduction of 26% and an increment of 21% have been observed during monsoon and post-monsoon respectively against their preceding seasons. The seasonal minimum and maximum spatial average TROPOMI-NO₂ level over the whole of Jharkhand averaged from 2019–2021 is summarised in Fig. S7. It has been observed that the winter exhibits the highest level of seasonal maximum TROPOMI-NO₂ (17.2×10^{15} molecules.cm⁻²), and pre-monsoon shows the highest level of seasonal minimum TROPOMI-NO₂ value (1.64×10^{15} molecules.cm⁻²), whereas monsoon exhibits the lowest level of both seasonal minimum and maximum TROPOMI-NO₂ values having 1.05×10^{15} molecules.cm⁻² and 7.28×10^{15} molecules.cm⁻² respectively.

In India, the maximum and minimum NO₂ level is reported during winter to pre-monsoon and monsoon respectively (Biswal et al., 2021). For the four seasons, a range of TROPOMI-NO₂ follows the order winter (15.8×10^{15} molecules.cm⁻²) > post-monsoon (15.4×10^{15} molecules.cm⁻²) > pre-monsoon (10.6×10^{15} molecules.cm⁻²) > monsoon (6.23×10^{15} molecules.cm⁻²) (Table S3). The frequency distribution of pixels that are loaded with seasonal mean TROPOMI-NO₂ for the four seasons winter, pre-monsoon, monsoon, and post-monsoon averaged from 2019 to 2021 in Jharkhand is depicted in Fig. S8 as the histogram plot. For all the four seasons, TROPOMI-NO₂ appears to be well-distributed and unimodal and is also found to be positively skewed. The overall count of pixels having low TROPOMI-NO₂ value is always observed to be large in number as compared to the count of pixels having high TROPOMI-NO₂ value, which clearly suggests that the whole of Jharkhand is mainly covered with the low TROPOMI-NO₂ pixels, and the burden of high TROPOMI-NO₂ is in some specific regions, which are captured and surrounded by the major industrialised and TPP locations such as Dhanbad and Jamshedpur (Bandyopadhyay et al., 2021). NO₂ is acidic and a highly reactive gas, so its lifetime is perceived to be short. That is why TROPOMI-NO₂ maps tell us only where does the NO₂ emission occur and not much about where

does the NO_2 blow, because at the end, it reacts out (Beirle et al., 2003; Beirle et al., 2011).

Evaluation of Relative difference in TROPOMI- NO_2 at the district level

The study is further extended by adding analysis and observation of annual mean spatial average TROPOMI- NO_2 variability at the district level of Jharkhand state represented in Fig. S9. As compared to 2019, in the year 2020, all the 24 districts experienced a decline in TROPOMI- NO_2 , which ranges from 4 to 16%; hence, none of these suffered from the elevated NO_2 level in 2020. Therefore, it is concluded that the COVID-19-induced lockdown not only contributed positively to reducing the TROPOMI- NO_2 level in major urban or industrial districts but also blessed other rural settlements too. It is attributed to the strict enforcement of control measures in Jharkhand in 2020 following the COVID-19 pandemic, resulting in a complete lockdown to prevent the spreading of the COVID-19 virus.

A drop of 20–40% in NO_2 level was observed in all major Indian cities with the exception of comparing it with the preceding year (Sur et al., 2021). The revival of normalcy increased the NO_2 level ranges from 10 to 34% in the post-COVID-induced lockdown year (2021) as compared to the COVID-induced lockdown year (2020), whereas an increment ranges from 1 to 23% and decrement by 3% and 2% in Garhwa and Simdega respectively while comparing with the pre-COVID-induced lockdown year (2019). Hence, it is obvious that the positive outcome of COVID-19-induced restrictions on TROPOMI- NO_2 level was not only temporary but also directly related to the emission of NO_2 . The interpretation clearly indicates that the major and most active industrial districts such as Dhanbad, Bokaro, and Purbi- Singhbhum regions are always found to have a high TROPOMI- NO_2 level during our study period (Bandyopadhyay et al., 2021).

Analysis of Spatiotemporal variation of NO_2 concentration at eight ground-monitoring stations over Jharkhand

Monthly averaged ground-based NO_2 and monthly averaged TROPOMI- NO_2 values were gathered for the selected monitoring stations for the periods 2017–2020 and 2019–2021 to investigate how concentration evolved over a period of time.

Analysis of NO_2 concentrations at point stations with the aid of a satellite approach

Month-on-month differences in TROPOMI- NO_2 level at all eight ground-monitoring stations in Jharkhand are illustrated in Fig. S10. In order to achieve this target, to start with, the GPS location of all eight considered JSPCB monitoring

stations was identified and located on the working raster dataset. Thereafter, with the aid of identify tool in ArcGIS software (ver. 10.4), the TROPOMI- NO_2 value is gathered for that particular GPS location which lies on their respective pixel of the TROPOMI- NO_2 raster dataset. However, the magnitude of temporal variation in NO_2 concentration is found to be different at all eight monitoring stations. The station-wise temporal variation of TROPOMI- NO_2 suggests that two monitoring stations Golmuri and Tata situated in the Jamshedpur district always exhibit a higher TROPOMI- NO_2 level as compared to the other six stations in Jharkhand. Both the stations follow almost the same pattern of variation, even though having a difference in the environment of adjoining areas. In the same way, the rest of the five stations Kusunda, Bastacola, Jharia, City Plaza, and Sindri located in Dhanbad display a nearly identical pattern of variation, despite having dissimilar territorial compositions.

These two districts are packed with all major industries, thermal power plants, and mining activities (Bandyopadhyay et al., 2021), which emit NO_2 into the atmosphere on a larger scale. Among all stations, Albert Ekka always displayed lower TROPOMI- NO_2 throughout the study period. This station-wise variation in NO_2 level may also be influenced by the level of effectiveness of pollution control measures enforced across local stations in Jharkhand. On one hand, during the concurrent COVID-19 year (2020), the TROPOMI- NO_2 levels observed were 4.94, 5.06, 5.20, 4.52, 4.62, 6.35, 9.14, and 1.84 ($\times 10^{15}$ molecules. cm^{-2}) during March month and 3.91, 4.07, 4.09, 3.55, 4.30, 4.17, 5.86, and 1.86 ($\times 10^{15}$ molecules. cm^{-2}) during April month at City Plaza, Bastacola, Kusunda, Jharia, Sindri, Tata, Golmuri, and Albert Ekka respectively. On the other hand, in the pre-COVID year (2019), the TROPOMI- NO_2 level was found to be 6.02, 6.17, 5.77, 5.33, 6.29, 8.03, 10.8, 2.43 ($\times 10^{15}$ molecules. cm^{-2}) during March month and 5.42, 5.50, 5.54, 4.96, 5.27, 5.27, 6.92, and 2.51 ($\times 10^{15}$ molecules. cm^{-2}) during April month at the above-mentioned stations respectively. Whereas, in the post-COVID year (2021), the TROPOMI- NO_2 levels were seen elevated up to 8.81, 9.54, 9.09, 7.77, 7.77, 10.1, 12.2, and 3.08 ($\times 10^{15}$ molecules. cm^{-2}) during March month and 5.73, 5.82, 5.42, 4.87, 5.74, 8.93, 11.0, and 2.37 ($\times 10^{15}$ molecules. cm^{-2}) during April month at the above-mentioned stations respectively.

Analyses of 3-year data show that the level of TROPOMI- NO_2 during the lockdown scenario (2020) decreased substantially as compared to the preceding (2019) and successive (2021) years during the same time period at all the considered stations in Jharkhand, which indicate a positive impact of COVID-19-induced measures leading to an ambient air. The Box-Whisker plot of TROPOMI- NO_2 at eight monitoring stations is plotted in Fig. S11, which excludes outliers at each of the monitoring sites. Box plot distribution also highlights that TROPOMI- NO_2 variability is highest at

Tata and Golmuri sites, and these two stations always face the burden of TROPOMI-NO₂, whereas Albert Ekka representing the lowest variability.

Analysis of NO₂ concentrations at point stations via in situ measurement

For the confirmation of these satellite observations as described in the earlier section ([Analysis of NO₂ concentrations at point stations with the aid of satellite approach](#)), to quantify the observation near the surface as well, ground-based measurements of trace gas NO₂ are also tracked. Based on data availability, month-wise variability of ground-NO₂ profiles at three different monitoring stations Tata, Golmurim and Albert Ekka are shown in Fig. S12 for considered years 2017–2020. The monthly mean surface-NO₂ data have been generated by averaging the available daily mean surface-NO₂ data for their respective monitoring station. Ground observation also highlighted the same scenario of variation for stations Golmuri and Tata that are located in heavily industrialised district of Jamshedpur, despite the dissimilar surrounding environment.

Ground-NO₂ concentrations are observed to be in the range of 45–53 µg/m³ during most of the time at these two stations, whereas the occurrence of the higher value of ground-NO₂ in Albert Ekka is very rare, and it reaches up to 46.7 µg/m³ only once in the month of October 2019, during the study period 2017–2020. At this station (Albert Ekka), most of the time ground-NO₂ concentrations are noted to be in the range of 35–38 µg/m³. As highlighted in Figs. 3 and S5, two stations Golmuri and Tata are always found to be in the hotspot (higher concentration) region represented as red colour in the southern region of Jharkhand, whereas Albert Ekka station does not lie in the hotspot area. One major finding is noticed here that a sharp decline of ground-NO₂ at two monitoring stations based out of Jamshedpur occurred soon after the implementation of total lockdown as we compare it with the same time period of the preceding (2019) year. As viewed from the graphical representation, the reduction is seen from 48.18 to 22.41 µg/m³ and from 47.34 to 22.03 µg/m³ for Tata and Golmuri respectively from March to April of the COVID-induced lockdown year (2020). A reduction of such magnitude happens to be lower than the NO₂ level in the rainy seasons of all three studied years. It shows that Tata and Golmuri stations witnessed a drastic decline in surface-NO₂, which may be called a historic event, not seen in the last two–three decades.

It may give us a sense that the COVID period acted as a short-term blessing for all from an air pollution perspective. As the COVID-19-induced lockdown had to come to an end, so does the improvement in air quality had to disappear. This huge difference in satellite- and ground-based measurement of NO₂ can be attributed to the fact that elevated

NO₂ concentration were found to be present in and around the sources of emissions such as thermal regions and urban and industrial centres, whereas less NO₂ is seen near the surface in a rural region and more in the upward atmosphere because of the gradual dilution and transportation due to meteorological factors. The observation made using both the approaches, i.e. ground-based monitoring and satellite-based interpretation, justifies the reduction in NO₂ concentration during lockdown and thus strengthens the findings of this study.

Meteorological variation along with their relationship with TROPOMI-NO₂ and detection of relative difference

The analyses of the pollution level of ambient air are found to be complicated due to meteorological influences. It plays an important part in air pollution formation, as well as its transportation, accumulation, and transformation too. An unfavourable meteorological situation could accentuate extreme pollution events despite a drop in total emission (Wang et al., 2019). Hence, its role is significant while determining air pollution levels. This section examines dependencies and the relation of monthly average TROPOMI-NO₂ with the four monthly average meteorological parameters, precipitation, temperature, relative humidity, and wind speed, which were carried out for all eight monitoring stations for the studied period 2019–2021. The captured correlation was found to be remarkable having moderate to strong relation in multiple cases (Table 1). Among all the four parameters, precipitation was found to be influenced more by having a strong negative correlation with TROPOMI-NO₂ ($R = 0.6–0.8$), which depicts that as the precipitation decreases, TROPOMI-NO₂ level increases, and vice-versa.

Along with the considered meteorological parameters, the time series analysis of TROPOMI-NO₂ at eight stations is graphically depicted in Fig. 4a–h. In the time series plot, when the observed meteorological parameters and TROPOMI-NO₂ level during the COVID-19-induced lockdown period were analysed against the pre- and post-COVID-19-induced lockdown period, it would be observed that both kept on changing and fluctuating according to their relationship seen in Table 1. As such, at Golmuri station, among all the four considered meteorological factors, precipitation appeared to be strongly negatively correlated with TROPOMI-NO₂ having $R = 0.8$. The time series analysis at Golmuri station clearly showed that in March month of 2020, precipitation was in the range of 2.22 mm/day, which is the less optimal condition for the dispersion of air pollutants, and also, TROPOMI-NO₂ was elevated (Fig. 4b). However, soon after the lockdown (24th March 2020), in April month, precipitation was in the range of 3.49 mm/day, which are a comparatively more favourable environment for a much better dispersion of air pollutants in the atmosphere.

Table 1 Representing correlation analysis for monthly mean TROPOMI-NO₂ (molecules.cm⁻²); scale 1×10^{15} along with the four Monthly mean meteorological parameters: (a) temperature (°C), (b) relative humidity (%), (c) wind speed (m/s), and (d) precipitation

(mm/day) for the eight ground-monitoring stations during 2019–2021, (a) City Plaza [D], (b) Tata [J], (c) Kusunda [D], (d) Jharia [D], (e) Sindri [D], (f) Bastacola [D], (g) Golmuri [J], (h) Albert Ekka [R], at 95% confidence level (CL)

Ground-monitoring stations	Pollutant	Meteorological parameters	Correlation (<i>R</i>)	<i>P</i> -value at 95% confidence level	Remarks
(a) City Plaza [D]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.5	0.004943	Statistically significant
		Relative humidity	<i>R</i> = −0.6	0.000283	Statistically significant
		Wind speed	<i>R</i> = −0.3	0.066305	Statistically not significant
		Precipitation	<i>R</i> = −0.6	2.24E-05	Statistically significant
(b) Tata [J]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.7	2.02E-07	Statistically significant
		Relative humidity	<i>R</i> = −0.4	0.038089	Statistically significant
		Wind speed	<i>R</i> = −0.7	6.27E-06	Statistically significant
		Precipitation	<i>R</i> = −0.8	2.69E-08	Statistically significant
(c) Kusunda [D]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.5	0.001672	Statistically significant
		Relative humidity	<i>R</i> = −0.6	0.000569	Statistically significant
		Wind speed	<i>R</i> = −0.3	0.0494914	Statistically significant
		Precipitation	<i>R</i> = −0.7	5.58E-06	Statistically significant
(d) Jharia [D]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.4	0.022086	Statistically significant
		Relative humidity	<i>R</i> = −0.7	2.13E-05	Statistically significant
		Wind speed	<i>R</i> = −0.2	0.18989129	Statistically not significant
		Precipitation	<i>R</i> = −0.7	1.76E-05	Statistically significant
(e) Sindri [D]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.6	0.000187	Statistically significant
		Relative humidity	<i>R</i> = −0.5	0.001363	Statistically significant
		Wind speed	<i>R</i> = −0.5	0.000644	Statistically significant
		Precipitation	<i>R</i> = −0.7	1.8E-06	Statistically significant
(f) Bastacola [D]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.5	0.003356	Statistically significant
		Relative humidity	<i>R</i> = −0.6	0.000213	Statistically significant
		Wind speed	<i>R</i> = −0.3	0.074692	Statistically not significant
		Precipitation	<i>R</i> = −0.7	1.07E-05	Statistically significant
(g) Golmuri [J]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.7	7.22E-07	Statistically significant
		Relative humidity	<i>R</i> = −0.5	0.004036	Statistically significant
		Wind speed	<i>R</i> = −0.7	7.148E-06	Statistically significant
		Precipitation	<i>R</i> = −0.8	1.415E-09	Statistically significant
(h) Albert Ekka [R]	TROPOMI-NO ₂	Temperature	<i>R</i> = −0.5	0.005181	Statistically significant
		Relative humidity	<i>R</i> = −0.4	0.00698	Statistically significant
		Wind speed	<i>R</i> = −0.5	0.005757	Statistically significant
		Precipitation	<i>R</i> = −0.7	9.2E-07	Statistically significant

Considering the same time zone at Golmuri station, precipitation was in the range of 1.21 mm/day and 2.34 mm/day in March and April months respectively in 2019, whereas in 2021, precipitation was in the range of 0.51 mm/day and 1.58 mm/day in March and April months respectively. Hence, TROPOMI-NO₂ was also found to be lifted up in the pre-COVID-induced lockdown period (2019) with 10.8 and 6.92 ($\times 10^{15}$ molecules.cm⁻²) in March and April months respectively and in post-COVID-induced lockdown period (2021) with 12.2 and 11 ($\times 10^{15}$ molecules.cm⁻²) as compared to the concurrent COVID-induced lockdown period (2020) with 9.14 and 5.86 ($\times 10^{15}$ molecules.cm⁻²) in the stated months respectively. Decrement and increment in TROPOMI-NO₂

and precipitation level respectively during the COVID period (2020) clearly highlights that in 2020, high precipitation leads to the rain washout which ultimately contributed to lowering TROPOMI-NO₂ level to some extent. Therefore, it is obvious that along with the influence of pandemic lockdown-induced restricted activities, meteorological factors also contributed and participated to some extent in the decline of TROPOMI-NO₂ levels in eight ground-monitoring stations, which agreed well with Jain and Sharma (2020), where they have analysed only three meteorological parameters, temperature, wind speed, and relative humidity for the megacities.

Nevertheless, these comparisons and observations over eight available stations appeared useful to see fluctuations

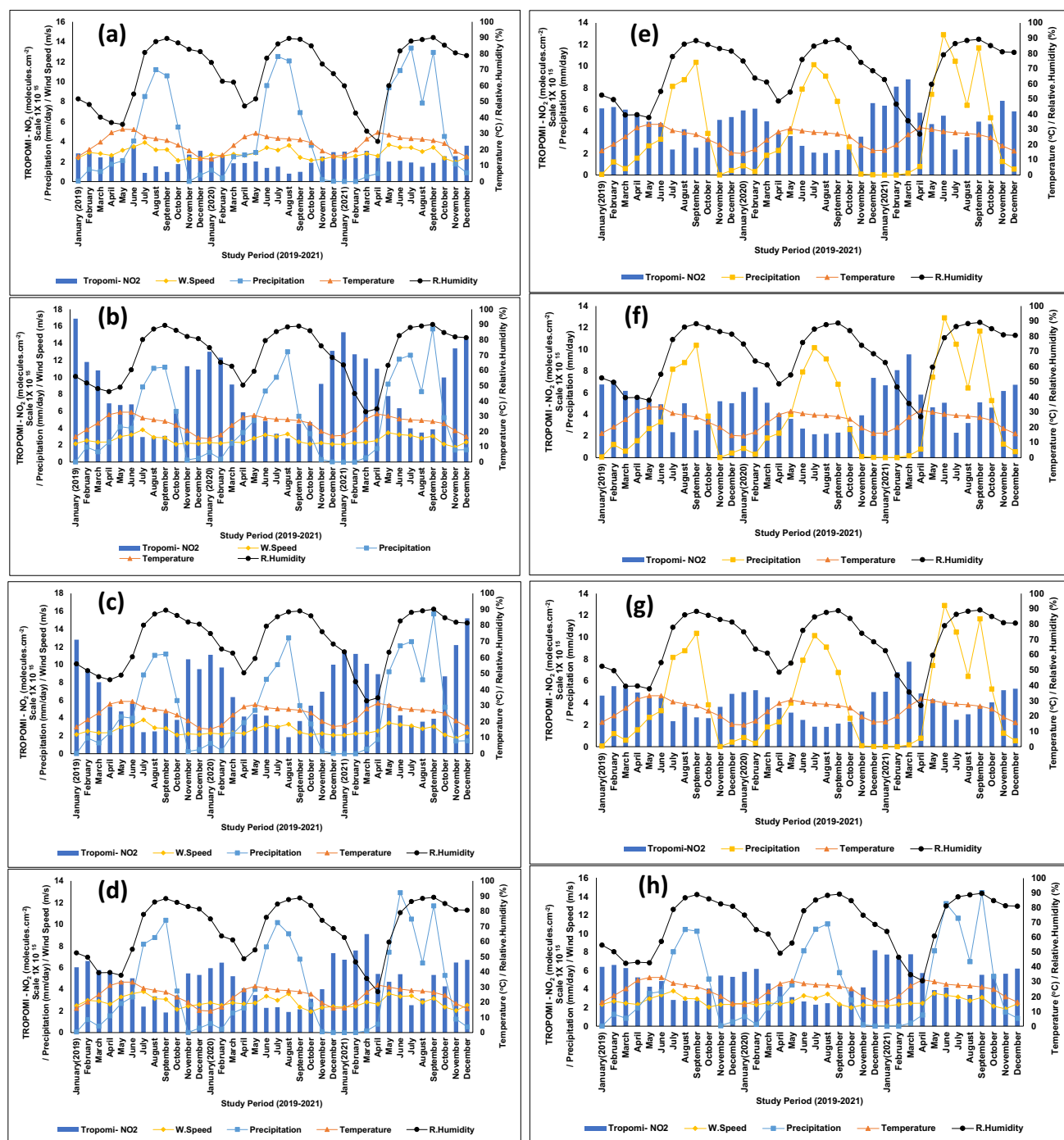


Fig. 4 Combo graph showing monthly average temperature (°C), relative humidity (%), precipitation (mm/day), wind speed (m/s), and TROPOMI-NO₂ (molecules.cm⁻²); scale 1x 10¹⁵ for eight ground-monitoring stations during 2019–2021, **a** Albert Ekka [R], **b** Golmuri

[J], **c** Tata [J], **d** Kusunda [D], **e** City Plaza [D], **f** Bastacola [D], **g** Jharia [D], **h** Sindri [D]. The R.H.S vertical line corresponds to the secondary vertical axis

and effects of meteorology locally. The present outcomes illustrate that as compared to 2019 and 2021, meteorological parameters were slightly more favourable in 2020 for deciding the TROPOMI-NO₂ level which is adroitly recorded by Goldberg et al. (2020), where they have observed

meteorological patterns to be especially optimal for having low NO₂ in most part of the USA during spring 2020, examined against spring 2019. It has explained well the effect of meteorology while deciding the level of TROPOMI-NO₂ is vital.

As we all know, 7 months in 2020 have gone through the pre-lockdown (1st January 2020–23rd March 2020), lockdown (24th March 2020–31st May), and unlock phases (1st June–31st July), which have been considered here for performing relative difference analysis by considering the same period in 2019. A clear view of TROPOMI-NO₂ levels during all the considered time frames for 2019 and 2020 and the existing relative difference over eight monitoring stations are summarised in Fig. S13a–h. The interpretation demonstrated that there exists a massive percentage drop in TROPOMI-NO₂ at all eight considered stations in 2020 compared to 2019. These percentage declines are different and varying for all stations, but the maximum decline seen is up to 28–30% and 26–30% found in the Jharia and Bastacola stations of Dhanbad respectively, which is a heavily mined and industrialised city. During the unlock period in 2020 as weighed against the same period of 2019, a drop in TROPOMI-NO₂ was also found to be in the higher range of up to 45% and 46% in Jharia and Bastacola respectively.

These huge percentage declines during the unlock period are mainly attributed to the dominant influence of the rain washout as having a strong negative correlation up to 0.7–0.8 between TROPOMI-NO₂ and two meteorological factor precipitation and R.H. These relations reveal that these two meteorological parameters have influenced significantly in deciding the level of TROPOMI-NO₂ during unlock period in 2020.

The time series analysis presented in Fig. 4g clearly highlights that precipitation was in the range of 3.29 mm/day and 7.91 mm/day whereas R.H was in the range of 54.88% and 75.88% during the June month of 2019 and 2020 respectively in Jharia. On the contrary, for April, precipitation was in the range of 1.55 mm/day and 2.26 mm/day whereas R.H was in the range of 39.62% and 48.62% in 2019 and 2020 respectively at Jharia. In a similar way, for the May month, precipitation was in the range of 2.68 mm/day and 3.98 mm/day whereas R.H was in the range of 37.75% and 54.5% in 2019 and 2020 respectively in Jharia. It clearly indicates that not much differences exist in meteorological parameters during the lockdown period. But there are vast differences in meteorological parameters while having unlock period. With that note, it is inferred that during the assessment period of complete lockdown, the observed percentage decline in TROPOMI-NO₂ is found to be directly related to and affected by the reduced human activities and industrial production. The present outcome is found to be consistent with the changes highlighted by Biswal et al. (2020), where they noticed a decrease of 20.9% in Jharkhand and in the range of up to 6.9 to 65.9% in other 27 states while also observed an increment of up to 3.7 to 39.1% in the remaining seven states of India, as compared with TROPOMI-NO₂ concentration during 2020 and pre-lockdown period (2019).

Assessment of TROPOMI-NO₂

TROPOMI-NO₂ is the columnar data that represents the density of NO₂ from the surface of the earth to the top of the troposphere or to the tropopause. To find out the consistency between satellite observations and surface observation of NO₂, research was carried out through the assessment and evaluation of Sentinel-5p. Further process was conducted by sorting the monthly mean surface-NO₂ data and then deriving seasonal data for the considered monitoring stations. The pronounced relationship between surface-NO₂ and TROPOMI-NO₂ is depicted graphically in Fig. 5a–d, which revealed that the model performance was found to be better during winter and post-monsoon compared to pre-monsoon and monsoon. In winter and post-monsoon seasons, good agreement is obtained between them with an $R = 0.8$, $R^2 = 0.71$ ($p < 0.01$), and $R = 0.71$, $R^2 = 0.50$ ($p < 0.01$) respectively, but huge discrepancies are seen between them during the pre-monsoon and monsoon, with the assessment of correlation $R = 0.45$, $R^2 = 0.20$ ($p < 0.01$), and $R = 0.16$, $R^2 = 0.03$ ($p > 0.05$) respectively. This is done to appraise whether the seasonal variability in the column emulates variability in the surface concentrations. The reason for not having an appreciable correlation during monsoon and pre-monsoon could be because of a reduction in the signal-to-noise ratio, possibly found to be associated with the drop in NO₂ concentration due to scavenging in monsoon (Pawar et al., 2017) and due to formation of secondary pollutants during pre-monsoon season (ul-Haq et al., 2018; Pawar et al., 2017).

Present outcomes are in tandem with similar work using the Sentinel-5p columnar tropospheric NO₂ and surface-NO₂ with the good and strong correlation ($R = 0.7$) between them (Bassani et al., 2021; Biswal et al., 2021) where they considered the lockdown phase for both years 2019 and 2020 for analysis in Rome and India respectively. Line plot and comparison highlight that surface-NO₂ shows trends identical to TROPOMI-NO₂ collected from the same location, during the winter season, and average for before and after the monsoon, whereas poor in monsoon Fig. S14a–d. Thus, it gives much confidence that TROPOMI-NO₂ can be a useful tool for monitoring surface-NO₂ in these two seasons.

Time series

Later on, a detailed analysis of NO₂ concentration during the complete lockdown phase (26th March–30th May) was executed by plotting the temporal variations of spatial 3-day average TROPOMI-NO₂ over the whole of Jharkhand state for 2019 and 2020 represented in Fig. 6. It was noted that both the years have decreasing trend which was also observed in Fig. S2 where the spatial monthly mean TROPOMI-NO₂ also drops as we move from March to May month. It has been clearly shown that immediately after

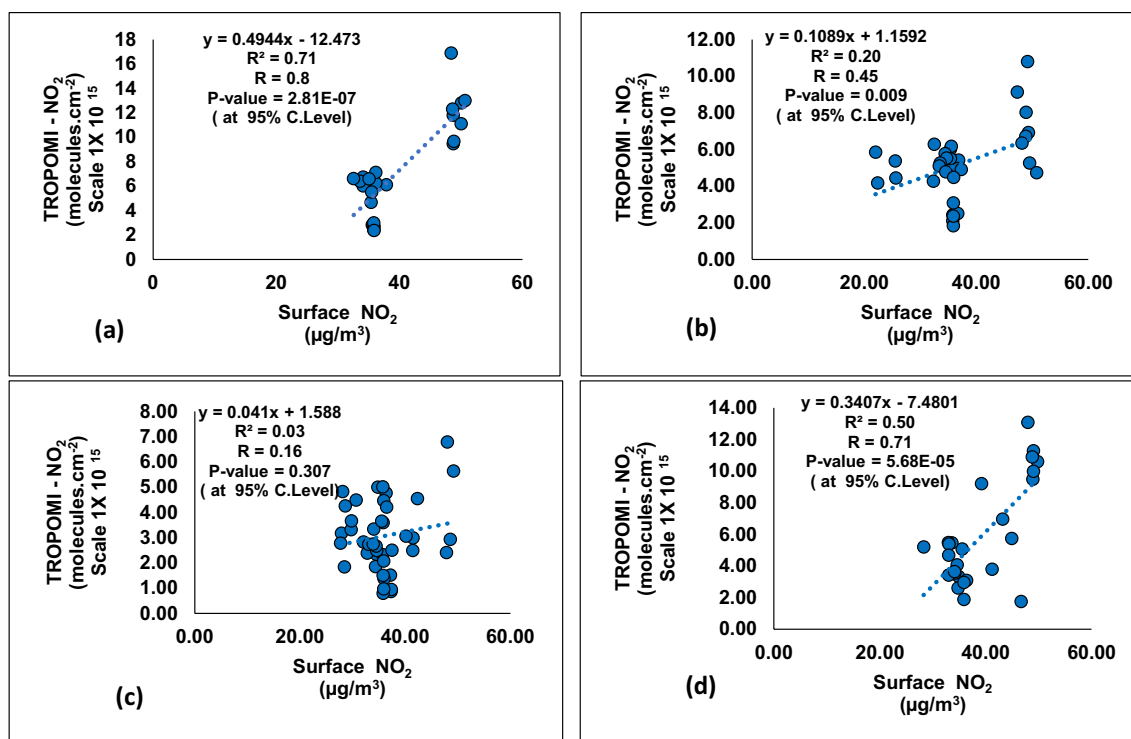
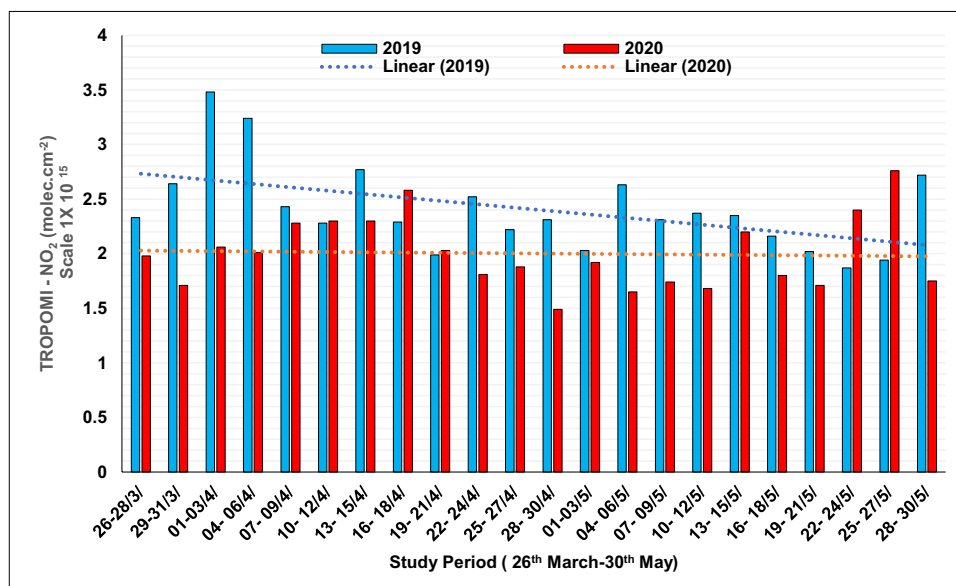


Fig. 5 a–d Scatter plots of monthly mean surface-NO₂ (µg/m³) versus monthly mean TROPOMI-NO₂ (molecules.cm⁻²); scale 1×10¹⁵ are represented for four seasons during 2019–2021: **a** winter, **b** pre-monsoon, **c** monsoon, and **d** post-monsoon

the total lockdown was imposed on 24th March, a sharp decline in TROPOMI-NO₂ appeared, which is represented by a negative slope (red colour line) in 2020, while in 2019, high values were observed throughout the study duration compared to 2020. In this graph, some peaks of TROPOMI-NO₂ release have also been captured for Jharkhand in 2019

whereas during the lockdown phase in 2020, it stabilised. Hence, a huge contrast was found through the lockdown periods in 2020 weighed against the same periods in 2019. Our result showed that Jharkhand experienced a positive outcome of lockdown in improving the TROPOMI-NO₂ level verified in Fig. 6. The result is consistent with other Indian

Fig. 6 Temporal variation in 3 days mean spatial averaged TROPOMI-NO₂ (molecules.cm⁻²) over Jharkhand state during 26th March to 30th April for both the years 2019 and 2020. The blue and red lines show a linear trend in the average concentration of spatial averaged TROPOMI-NO₂ during 2019 and 2020



cities that are Kolkata and Mumbai studied by R. P. Singh and Chauhan (2020) for the month of March to the first week of April for 2019 and 2020, where they have not considered the May month, which formed part of the lockdown phase.

Conclusions

One of the consequences of air pollution happened to be a global health crisis, which needs to be resolved at the earliest. The present research aimed at studying the impact of COVID-19-induced lockdown in terms of spatial and temporal variation of TROPOMI-NO₂ and ground NO₂ concentration in Jharkhand. It is concluded that a pronounced decline of 11% in TROPOMI-NO₂ has been identified in Jharkhand during the lockdown period. Among eight ground-monitoring stations, Jharia and Bastacola in Dhanbad experienced an enormous drop of up to 30% in TROPOMI-NO₂ during lockdown phase 2020 compared to 2019. This study demonstrated very well the role of understanding the contribution of meteorological conditions when designing emission control strategies. As in this case, it is found to be inversely proportional throughout the considered stations having $R = 0.4\text{--}0.7$, $0.4\text{--}0.7$, $0.2\text{--}0.7$, and $0.6\text{--}0.8$ for temperature, relative humidity, wind speed, and precipitation respectively.

Assessment of satellite observations against ground-based appraisal is noted to be in harmony with $R = 0.8$ and $R = 0.71$ during winter and post-monsoon respectively. Analyses found a tremendous fall in surface-NO₂ at Tata and Golmuri stations from 48.18 to 22.41 µg/m³ and from 47.34 to 22.03 µg/m³ respectively from March to April month of lockdown year (2020), which is always abundant with highly polluted air. At the district level, reduction ranges from 4 to 16% in 2020 compared to 2019 and staging a revert, ranging from 10 to 34% in 2021 as compared to 2020. Overall examination conducted via both satellite and ground observations highlighted that Jharkhand definitely has gone through the experience of remarkable improvement in NO₂ level during the COVID-19-led lockdown period.

The improved algorithm and more precise study with more ground-monitoring points will help in establishing a strong relationship between satellite data and ground-level NO₂ which will help in estimating NO₂ concentration in remote locations more precisely. The present outcomes could also aid in planning and implementing better air pollutant reduction strategies and upgrading all required air quality measurement and prediction modelling for the betterment of the whole universe. The use of advanced technologies in industries especially thermal power plants and vehicles will reduce the NO₂ emissions. The change in domestic fuels from biomass to cleaner fuel will further improve the overall air quality.

As we have gone through many more analysis and discussions, owing to limitations, we could not broaden our span of study, parameters, and tools. Meteorological behaviour and its role on the level of TROPOMI-NO₂ fluctuation have been analysed on sites located in the Central and North-eastern plateau and South-eastern plateau whereas similar analyses have not been done for the Western plateau. In upcoming research, interpretation and analysis may include a few more parameters such as boundary layer height or may apply some other approaches such as machine learning in order to extract profound insights about the studied domain. All the highlights may be targeted for future research to deal with the current situation of Jharkhand which is burdened with many industries and TPPs and a lot more in the pipeline.

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Authors' contribution All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by Shalini Priya and Jawed Iqbal. The first draft of the manuscript was written by Shalini Priya, and reviewing and editing was completed by Jawed Iqbal. All authors read and approved the final manuscript.

Data availability The secondary data obtained from JSPCB, Jharkhand, is available with the corresponding author and can be available on request. There is no other associated data.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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Authors and Affiliations

Shalini Priya¹  · Jawed Iqbal¹ 

✉ Jawed Iqbal
jawed@bitmesra.ac.in
Shalini Priya
phdcee10052.19@bitmesra.ac.in

¹ Department of Civil and Environmental Engineering, Birla Institute of Technology Mesra, Ranchi, Jharkhand 835215, India